CHAPTER

04

Heat and Thermodynamics

Thermodynamics is the science of the relationships between **heat, work, temperature and energy**. It describes how energy moves from one system to another and changes from one form to another. A **system** is the definite quantity of matter we study (for example, a gas in a cylinder with a piston); everything around it is the **surroundings**. A system's state is described by its **temperature, pressure, volume and composition**. This chapter moves from temperature, through expansion and calorimetry, to heat transfer and the **first law of thermodynamics**.

LESSONS IN THIS CHAPTER

4.1 Temperature and the Zeroth Law of Thermodynamics

4.2 Thermal Expansion

4.3 Heat, Work, Internal Energy; Specific and Latent Heat

4.4 Heat Transfer Mechanisms

4.5 The First Law of Thermodynamics

Freshman A+ · General Physics (Phys 1011)

CHAPTER 4 · LESSON 4.1

Temperature and the Zeroth Law of Thermodynamics

AFTER THIS LESSON YOU CAN

- Define thermal equilibrium and state the zeroth law
- Distinguish heat, temperature and internal energy
- Convert between the Celsius, Fahrenheit and Kelvin scales
- Build a conversion for any linear scale

4.1.1 Thermal equilibrium and the zeroth law

D THERMAL EQUILIBRIUM

Two objects are in **thermal equilibrium** when, in thermal contact, **no net energy flows** between them. Objects in thermal equilibrium have the **same temperature**.

D ZEROTH LAW OF THERMODYNAMICS

If objects A and B are each in thermal equilibrium with a third object C, then A and B are in thermal equilibrium with each other.

It is "zeroth" because it was recognised (by R. Fowler, 1930s) after the first, second and third laws had been numbered, yet it logically comes first: it is what makes a **thermometer** possible.

Temperature is *the property that determines whether an object is in thermal equilibrium with other objects*. At the microscopic level, it measures the **average kinetic energy** of the random motion of the particles.

★ HEAT

- **Energy in transit** caused by a **temperature difference**.
- Flows spontaneously from **hot to cold**.
- A body **never contains heat**; heat exists only while crossing a boundary.
- Unit: joule (J); 1 cal = 4.186 J.

E TEMPERATURE AND INTERNAL ENERGY

- **Temperature**: how hot (average KE per particle); measured in °C or K.
- **Internal energy** U: the **total** microscopic energy of all the particles.
- A bathtub of warm water has **more internal energy** than a cup of boiling water, even though the cup is at a **higher temperature**.

4.1.2 Temperature scales

Any property that changes reproducibly with temperature (the volume of mercury or alcohol, electrical resistance, colour, infrared emission) can be used to build a thermometer. Scales are fixed by two reference points, usually the **ice point** and **steam point** of water at 1 atm.

Scale	Ice point	Steam point	Divisions between them
Celsius	0 °C	100 °C	100
Fahrenheit	32 °F	212 °F	180
Kelvin	273.15 K	373.15 K	100

CONVERSIONS

$$T_F = \tfrac{9}{5}T_C + 32 \qquad T_C = \tfrac{5}{9}(T_F - 32) \qquad T = T_C + 273.15$$

- A temperature **change** of 1 °C = 1 K = 1.8 °F.
- **Absolute zero** = 0 K = −273.15 °C: the **Kelvin scale has no negative values**.
- −40 °C = −40 °F (the two scales cross there).
- The SI unit of temperature is the **kelvin**, defined from the **triple point of water** (273.16 K).

ANY LINEAR SCALE X: THE RATIO METHOD

$$\frac{T_X - X_{\text{ice}}}{X_{\text{steam}} - X_{\text{ice}}} = \frac{T_C - 0}{100 - 0}$$

Example (module activity): ice point 50 °X, steam point 200 °X. For 60 °C: $\frac{T_X - 50}{150} = \frac{60}{100} \Rightarrow T_X = 140$ °X.

WORKED EXAMPLE 1: CONVERSIONS (ARBA MINCH AND WOLAITA SODO EXAMS)

(a) Convert 253 K to °F. (b) Convert 65 °F to °C.

(a) $T_C = 253 - 273 = -20$ °C → $T_F = \frac{9}{5}(-20) + 32 = -4$ °F.

(b) $T_C = \frac{5}{9}(65 - 32) = 18.3$ °C.

TEMPERATURE TRAPS

- "The hotter object contains more heat" is **wrong** on two counts: objects contain **internal energy**, not heat, and internal energy also depends on **mass**.
- Objects in thermal equilibrium have the same temperature and **no net** energy transfer.
- For a temperature **difference**, add or subtract 273 is not needed: ΔT in K equals ΔT in °C.

Exam Practice

8 QUESTIONS · LESSON 4.1

1 Wolaita Sodo University · Final Exam 2014

If two objects are in thermal equilibrium, they have the same temperature and there is a net energy transfer between them.

True / False

2 Wolaita Sodo University · Final Exam 2014

"When two systems are each in thermal equilibrium with a third system, they are in thermal equilibrium with each other." This is the:

A) First law of thermodynamics
B) Second law of thermodynamics
C) Mechanical equivalent of heat
D) Zeroth law of thermodynamics

3 Wolaita Sodo University · Final Exam 2014

Which of the following temperature scales has no negative values?

A) Celsius
B) Kelvin
C) Réaumur
D) Fahrenheit

4 Arba Minch University · Final Exam

The temperature on the Fahrenheit scale corresponding to 253 K is:

A) 4 °F
B) 12 °F
C) −4 °F
D) −36 °F

5 Wolaita Sodo University · Final Exam 2014

If the temperature of a body is 65 °F, what is it in °C?

6 Wolaita Sodo University · Final Exam 2014

Fill in the blank: "_____ is the property that determines whether an object is in thermal equilibrium with other objects."

7 Model question (module activity)

On an unknown scale X, the ice point is 50 °X and the steam point is 200 °X. What does a thermometer on scale X read at 60 °C?

8 Serway / OpenStax-style problem

At what temperature do the Celsius and Fahrenheit scales give the same reading?

A) 0°
B) −32°
C) 40°
D) −40°

Answers & Explanations

1. **False**

In thermal equilibrium they have the same temperature and **no** net energy flows between them.

2. **D**

Zeroth law of thermodynamics

This transitivity statement is the zeroth law, the basis of temperature measurement.

3. B

Kelvin

The Kelvin scale starts at absolute zero (0 K), the lowest possible temperature.

4. C

−4 °F

253 K = −20 °C; $T_F = 1.8(-20) + 32 = -4$ °F.

5. 18.3 °C

$T_C = \frac{5}{9}(65 - 32) = \frac{5}{9}(33) = 18.3$ °C.

6. Temperature

Two objects are in thermal equilibrium exactly when their temperatures are equal.

7. 140 °X

$$\frac{T_X - 50}{200 - 50} = \frac{60}{100} \Rightarrow T_X = 50 + 90 = 140 \text{ °X}.$$

8. D

−40°

Set $T = \frac{9}{5}T + 32 \Rightarrow -\frac{4}{5}T = 32 \Rightarrow T = -40$.

CHAPTER 4 · LESSON 4.2

Thermal Expansion

AFTER THIS LESSON YOU CAN

- Explain why materials expand when heated
- Use the formulas for linear, area and volume expansion
- Relate α, β and γ
- Explain bimetallic strips and the anomalous expansion of water

4.2.1 Why things expand

When a solid is heated, its atoms vibrate with larger amplitude, so the **average distance between them increases** and the object grows in every dimension. Hot air expands and becomes **less dense**, so it rises (smoke, hot-air balloons).

4.2.2 Linear, area and volume expansion

EXPANSION FORMULAS

Type	Formula	Coefficient
Linear (length)	$\Delta L = \alpha L_0 \, \Delta T$	α, unit (°C)$^{-1}$ or K^{-1}
Area	$\Delta A = \beta A_0 \, \Delta T$	$\beta = 2\alpha$
Volume	$\Delta V = \gamma V_0 \, \Delta T$	$\gamma = 3\alpha$
New size: $L = L_0(1 + \alpha \Delta T)$, and similarly for area and volume.		

RELATIONS BETWEEN THE COEFFICIENTS

$$\beta = 2\alpha, \qquad \gamma = 3\alpha, \qquad \gamma = \tfrac{3}{2}\beta$$

So if metal A has three times the linear coefficient of metal B, it also has **three times** the area coefficient ($\beta_A = 2\alpha_A = 6\alpha_B = 3\beta_B$). Liquids and gases have only a volume coefficient.

Typical values: steel $\alpha \approx 11$–12×10^{-6} °C^{-1}, aluminium 24×10^{-6}, brass 19×10^{-6}, glass 9×10^{-6}, Pyrex 3.2×10^{-6}.

WORKED EXAMPLE 1: RAILWAY TRACK

A steel rail ($\alpha = 12 \times 10^{-6}$ °C^{-1}) is 30.0 m long at 0 °C. How much longer is it at 40 °C?

$\Delta L = \alpha L_0 \Delta T = (12 \times 10^{-6})(30.0)(40) = 1.44 \times 10^{-2}$ m = 1.44 cm. This is why rails, bridges and concrete roads have **expansion gaps**.

WORKED EXAMPLE 2: A HOLE IN A PLATE

A circular hole in an aluminium plate is 2.000 cm across at 20 °C. What is its diameter at 220 °C? ($\alpha = 24 \times 10^{-6}$ °C⁻¹)

A hole **expands as if it were made of the material**: $d = 2.000(1 + 24 \times 10^{-6} \times 200) = 2.0096$ cm.

4.2.3 Applications

E BIMETALLIC STRIP

Two metals with **different α** are bonded together. When heated, the one with the **larger α expands more**, so the strip **bends toward the metal with the smaller α**. Used in **thermostats**, fire alarms, and dial thermometers.

E ANOMALOUS EXPANSION OF WATER

Water **contracts** when heated from 0 °C to **4 °C**, and expands above 4 °C. So water has its **maximum density at 4 °C**. Lakes freeze from the **top down**: the 4 °C water sinks, ice floats on top and insulates the water below, so fish survive the winter.

! EXPANSION TRAPS

- A **hole** in a heated plate gets **bigger**, not smaller.
- $\gamma = 3\alpha$, not α^3; and $\gamma = \frac{3}{2}\beta$.
- Use the temperature **change** ΔT; °C and K give the same number.
- When a full glass container of liquid is heated, liquid **overflows** because liquids usually have a larger γ than the glass.

Exam Practice

6 QUESTIONS · LESSON 4.2

1 Wolaita Sodo University · Final Exam 2014

The correct relationship between the area expansion coefficient β and the volume expansion coefficient γ is:

A) γ = 3β
B) γ = β/3
C) γ = 2β
D) γ = 2β/3
E) γ = 3β/2

2 Model question (module activity)

Metal A has a linear expansion coefficient three times that of metal B. Their area expansion coefficients compare as:

A) $\beta_A = \beta_B$
B) $\beta_A = 9\beta_B$
C) $\beta_A = 6\beta_B$
D) $\beta_A = 3\beta_B$

3 Serway / OpenStax-style problem

A brass strip ($\alpha = 19 \times 10^{-6}$ °C⁻¹) is bonded to a steel strip ($\alpha = 11 \times 10^{-6}$ °C⁻¹). When the strip is heated it bends:

A) toward the steel side
B) toward the brass side
C) it does not bend
D) first one way then the other

4 Model question

A metal plate with a circular hole in it is heated. The diameter of the hole:

A) increases
B) stays the same
C) decreases
D) decreases then increases

5 Serway / OpenStax-style problem

A copper rod ($\alpha = 17 \times 10^{-6}$ °C⁻¹) is 1.50 m long at 20 °C. Find its length at 120 °C, and its fractional change in volume.

6 Model question

Water has its maximum density at 0 °C.

True / False

Answers & Explanations

1. **E**

γ = 3β/2

$\beta = 2\alpha$ and $\gamma = 3\alpha$, so $\gamma/\beta = 3/2$.

2. **D**

$\beta_A = 3\beta_B$

$\beta = 2\alpha$: $\beta_A = 2(3\alpha_B) = 3(2\alpha_B) = 3\beta_B$.

3. **A**

toward the steel side

The brass expands more, so it ends up on the outside (longer) of the curve, and the strip curves toward the steel.

4. **A**

increases

Every dimension of the plate scales up by $(1 + \alpha\Delta T)$, including the hole, which expands as if it were filled with metal.

5. **1.50255 m; 5.1 × 10⁻³ (0.51 %)**

$\Delta L = (17 \times 10^{-6})(1.50)(100) = 2.55 \times 10^{-3}$ m → $L = 1.50255$ m.

$\Delta V/V = 3\alpha\Delta T = 3(17 \times 10^{-6})(100) = 5.1 \times 10^{-3}$.

6. **False**

Water is densest at **4 °C**. Between 0 °C and 4 °C it contracts on heating (anomalous expansion), and ice is less dense than water.

CHAPTER 4 · LESSON 4.3

Heat, Work, Internal Energy; Specific and Latent Heat

AFTER THIS LESSON YOU CAN

- Define heat, work and internal energy
- Use Q = mcΔT and heat capacity
- Use Q = mL for phase changes
- Solve calorimetry (mixing) problems including phase changes

4.3.1 Heat, work and internal energy

D THREE ENERGY IDEAS

- **Heat Q:** a **spontaneous** transfer of energy caused by a temperature difference. It is microscopic (many particle collisions); there is no large-scale force or displacement.
- **Work W:** a **non-spontaneous** transfer of energy by a force acting through a displacement (moving a piston, stirring, running a current through a resistor).
- **Internal energy U:** the energy of the **random, disordered motion** of the atoms and molecules: translational, rotational and vibrational kinetic energy plus the potential energy within and between molecules. The bulk kinetic energy of the whole system moving through space is **not** included.
  A system **has** internal energy; it does **not** have heat or work. Heat and work are two ways of **changing** U.

4.3.2 Specific heat and heat capacity

$$Q = mc\,\Delta T \qquad Q = C\,\Delta T,\quad C = mc \qquad Q = nC_m\Delta T \text{ (gases, per mole)}$$

- **Specific heat capacity** c: the heat needed to raise **1 kg** of a substance by **1 °C**. Unit: J/(kg·°C).
- **Heat capacity** C: the heat needed to raise the **whole object** by **1 °C**. Unit: J/°C.

★ WATER'S LARGE SPECIFIC HEAT

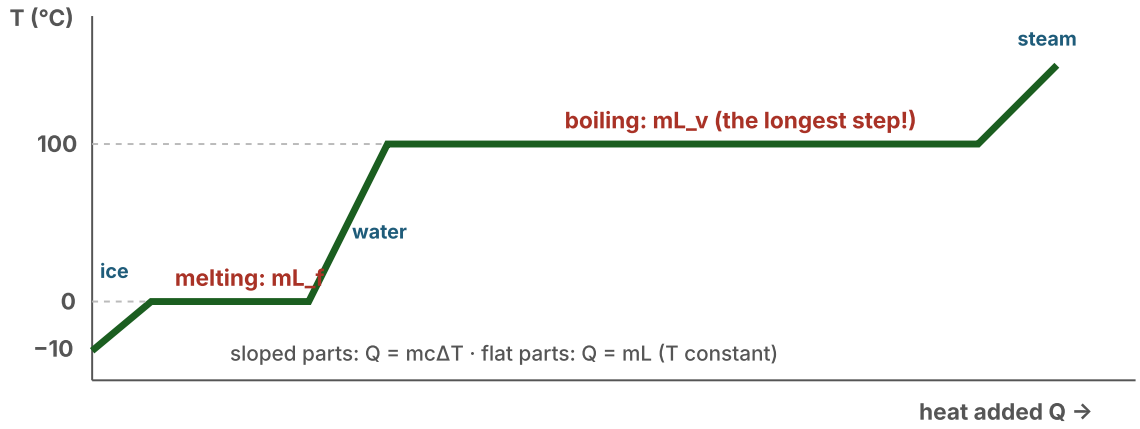
$c_{\text{water}} = 4186$ J/(kg·°C) = 1 cal/(g·°C), much larger than most substances (sand ≈ 800, copper 387, iron 448, aluminium 900, silver 234, lead 128). So in the sun **sand heats up much faster** than water, coastal climates are mild, and **water is an excellent fire extinguisher**: it absorbs a lot of heat per degree, and even more when it boils (large L_v).

4.3.3 Latent heat and phase changes

During a **phase change** (melting, boiling) the temperature **stays constant**: the heat breaks bonds instead of increasing the kinetic energy of the particles.

$$Q = mL$$

Latent heat	Change	Water
Fusion L_f	solid ↔ liquid	3.33×10^5 J/kg (80 cal/g)
Vaporization L_v	liquid ↔ gas	2.26×10^6 J/kg (540 cal/g)



Heating curve of water from ice at −10 °C to steam above 100 °C

WORKED EXAMPLE 1: ICE TO STEAM (MODULE)

How much heat changes a 40 g ice cube at −10 °C to steam at 110 °C? ($c_{\text{ice}} = 2090$, $c_w = 4186$, $c_{\text{steam}} = 2010$ J/(kg·°C))

Step	Calculation	Q (J)
Warm ice −10 → 0 °C	$0.040 \times 2090 \times 10$	836
Melt at 0 °C	$0.040 \times 3.33 \times 10^5$	13 320
Warm water 0 → 100 °C	$0.040 \times 4186 \times 100$	16 744
Boil at 100 °C	$0.040 \times 2.26 \times 10^6$	90 400
Warm steam 100 → 110 °C	$0.040 \times 2010 \times 10$	804
Total		**1.22 × 10⁵ J** (≈ 29.2 kcal)

Boiling alone takes about 74 % of the total.

4.3.4 Calorimetry: mixing problems

HEAT LOST = HEAT GAINED

In an insulated system, energy is conserved:

$$\sum Q_{\text{lost by hot}} = \sum Q_{\text{gained by cold}} \qquad \text{or} \qquad \sum Q = 0$$

Method: (1) check whether any phase change happens; (2) write each heat term with ΔT = (larger − smaller) so every term is positive; (3) set lost = gained; (4) check the final temperature makes sense.

WORKED EXAMPLE 2: MOLTEN LEAD IN AN IRON MOULD (MODULE)

90 g of molten lead at its melting point 327.3 °C is poured into a 300 g iron mould at 20.0 °C. Find the final temperature. (L_f lead $= 2.45 \times 10^4$ J/kg, $c_{Pb} = 128$, $c_{Fe} = 448$ J/(kg·°C))

Lead releases: $0.090(2.45 \times 10^4) + 0.090(128)(327.3 - T) = 2205 + 11.52(327.3 - T)$

Iron absorbs: $0.300(448)(T - 20.0) = 134.4(T - 20.0)$

$2205 + 3770 - 11.52T = 134.4T - 2688 \Rightarrow 145.9T = 8663 \Rightarrow T = 59.4$ °C.

✎ WORKED EXAMPLE 3: STEAM MEETS ICE (MODULE PROBLEM; A CLASSIC TRAP)

20 g of steam at 100 °C is added to 60 g of ice at 0 °C. Find the final temperature.

Heat available if **all** steam condenses: $0.020 \times 2.26 \times 10^6 = 45\,200$ J.

Heat needed to melt all the ice **and** warm it to 100 °C: $0.060 \times 3.33 \times 10^5 + 0.060 \times 4186 \times 100 = 19\,980 + 25\,116 = 45\,096$ J.

Condensation alone almost exactly does the job, so the mixture ends at **100 °C**, with all the ice melted and a tiny amount ($\approx (45\,200 - 45\,096)/2.26 \times 10^6 \approx 0.05$ g) of steam left.

If you blindly write one equation you get $T \approx 100.3$ °C, which is impossible: that is the signal to check phases.

! HEAT TRAPS

- During melting or boiling, the **temperature does not change** even though heat is added.
- Use **kg** with c in J/(kg·°C) and L in J/kg.
- Steam at 100 °C burns far worse than water at 100 °C, because it first releases L_v as it condenses.
- Heat capacity (J/°C) belongs to an **object**; specific heat (J/(kg·°C)) belongs to a **material**.

Exam Practice

8 QUESTIONS · LESSON 4.3

1 Wolaita Sodo University · Final Exam 2014

Fill in the blank: "_____ is defined as the amount of heat energy required to raise the temperature of a substance by 1 °C."

2 Wolaita Sodo University · Final Exam 2014

The thermal energy required to change 1 kg of a substance between liquid and gas without a change in temperature is called:

A) Latent heat of fusion
B) Latent heat of vaporization
C) Heat capacity
D) Specific heat capacity

3 Model question

During melting, the temperature of ice rises steadily as heat is added.

True / False

4 Model question (module problem)

The temperature of a 525 g silver bar rises by 10.0 °C when it absorbs 1.23 kJ of heat. Find the specific heat of silver.

5 Model question (module problem)

A 50.0 g copper sample at 25.0 °C absorbs 1200 J of heat ($c = 387$ J/(kg·°C)). Find its final temperature.

6 Model question (module problem)

How much heat is needed to change 1.0 g of ice at −10 °C into steam at 110 °C?

7 Model question (module problem)

20 g of steam at 100 °C is added to 60 g of ice at 0 °C in an insulated container. Find the final temperature.

8 Serway / OpenStax-style problem

Equal masses of sand and water absorb the same amount of heat from the sun. Which statement is correct?

A) Both reach the same temperature

B) The water gets hotter because it absorbs more heat

C) The sand gets hotter because it has a smaller specific heat

D) The sand gets hotter because it has a larger specific heat

Answers & Explanations

1. **Heat capacity**

$C = Q/\Delta T$ for the whole object. Per kilogram, it becomes the specific heat capacity c.

2. **B**

Latent heat of vaporization

Liquid ↔ gas at constant temperature = vaporization (or condensation). Fusion is solid ↔ liquid.

3. **False**

During a phase change the temperature stays constant; the heat goes into breaking bonds ($Q = mL_f$).

4. **234 J/(kg·°C)**

$$c = \frac{Q}{m\Delta T} = \frac{1230}{0.525 \times 10.0} = 234 \text{ J/(kg·°C)}.$$

5. **87.0 °C**

$$\Delta T = \frac{Q}{mc} = \frac{1200}{0.0500 \times 387} = 62.0 \text{ °C} \rightarrow T = 87.0 \text{ °C}.$$

6. **About 3.05×10^3 J**

Scale the 40 g example by 1/40: $(836 + 13\,320 + 16\,744 + 90\,400 + 804)/40 = 122\,104/40 = 3.05 \times 10^3$ J (about 730 cal).

7. **100 °C**

Condensing all the steam releases 45 200 J; melting all the ice and warming it to 100 °C needs 45 096 J. The heat available is (just) enough, so the system ends at 100 °C with a trace of steam left.

8. **C**

The sand gets hotter because it has a smaller specific heat

$\Delta T = Q/(mc)$: a smaller c gives a larger temperature rise.

CHAPTER 4 · LESSON 4.4

Heat Transfer Mechanisms

AFTER THIS LESSON YOU CAN

- Explain conduction, convection and radiation
- Use the conduction and radiation rate formulas
- Explain direct burning and the spread and control of fire
- Explain the vacuum flask

4.4.1 Conduction

Heat passes **from particle to particle** as they vibrate and collide; the particles themselves do not travel. It is most important in **solids**. **Metals** (silver, copper, aluminium) are the best conductors because their **free electrons** carry energy quickly. Most liquids (except mercury) and all gases are poor conductors (insulators).

$$P = \frac{Q}{t} = \frac{kA\,(T_H - T_C)}{L}$$

where k is the **thermal conductivity**, A the area and L the thickness.

Fire safety: fire can spread by conduction along **steel beams** and through **metal doors and handles**; an insulator wrapped around a heat source (a faulty electric blanket under pillows) traps heat until the ignition temperature is reached.

4.4.2 Convection

Heat is carried by the **bulk movement of the heated fluid itself**. Heated fluid expands, becomes **less dense** and **rises**; cooler, denser fluid sinks to take its place, setting up a **convection current**. It occurs only in **liquids and gases** (particles in a solid cannot move freely).

Examples: sea and land breezes, boiling water in a pot, room heaters, hot gases and burning fragments rising up **stairwells and lift shafts** in a building fire (while fresh air rushes in to feed the fire).

4.4.3 Radiation

Energy carried by **electromagnetic waves** (mainly infrared). It **needs no medium**: it travels through **vacuum**, which is how the Sun's heat reaches us. Radiation falling on a body may be **absorbed, reflected or transmitted**.

$$P = e\sigma A T^4, \qquad \sigma = 5.67 \times 10^{-8}\ \mathrm{W/(m^2K^4)}$$

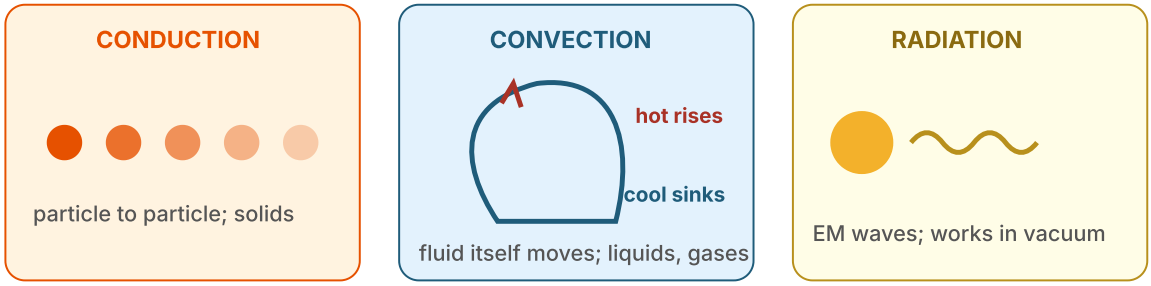
T must be in **kelvin**; e is the emissivity (0 to 1).

- **Black, dull surfaces** are good absorbers **and** good emitters; **white, shiny surfaces** reflect (hence the silver coating on a fire-fighter's suit).
- From a small source, the radiation received falls with the **square of the distance**: twice as far → ¼ of the energy.

- Glass transmits radiant heat without heating up much; a magnifying lens can focus sunlight enough to start a fire.

4.4.4 Direct burning

Some fire agencies list a fourth mechanism: **direct burning**, where a **flame touches** other fuel and spreads along a line of fuel (wood in contact with wood, a candle flame touching a curtain).



The three main heat-transfer mechanisms

E THE VACUUM (THERMOS) FLASK

It fights all three mechanisms: the **vacuum** between the double walls stops **conduction and convection** (there are no particles); the **silvered** walls reflect **radiation** back; the stopper reduces convection and evaporation at the top. So hot drinks stay hot and cold drinks stay cold.

! HEAT-TRANSFER TRAPS

- Only **radiation** works through a vacuum.
- Convection needs a **fluid** and **gravity** (density differences make warm fluid rise).
- A good absorber is also a good **emitter**.
- In $P = e\sigma AT^4$, doubling the **kelvin** temperature multiplies the power by 16.

Exam Practice

6 QUESTIONS · LESSON 4.4

1 Wolaita Sodo University · Final Exam 2014

_____ is a mechanism of heat transfer through vacuum.

A) Convection **B)** Conduction
C) Radiation **D)** Refraction

2 Wolaita Sodo University · Final Exam 2014

_____ is the transfer of heat by the bulk movement of the heated particles themselves.

A) Convection **B)** Conduction
C) Radiation **D)** Direct burning

3 Model question

Metals are good conductors of heat mainly because:

A) they are dense
B) they are shiny
C) their atoms are large
D) they have free electrons

4 Model question (module)

A fire-fighter's protective jacket has a shiny silver coating because shiny surfaces:

A) reflect radiant heat
B) conduct heat away quickly
C) absorb radiant heat well
D) stop convection

5 Model question (module)

An object is moved to twice its original distance from a small radiant heater. The radiant energy it receives becomes:

A) half
B) the same
C) twice
D) one quarter

6 Serway / OpenStax-style problem

A glass window is 1.0 m × 2.0 m and 4.0 mm thick ($k = 0.80$ W/(m·°C)). The inside surface is at 20 °C and the outside at 5.0 °C. Find the rate of heat loss by conduction.

Answers & Explanations

1. C

Radiation

Radiation is electromagnetic and needs no medium. Conduction and convection need particles.

2. A

Convection

In convection the heated fluid itself moves (rises), carrying energy with it.

3. D

they have free electrons

Free (conduction) electrons move through the metal and transfer energy quickly.

4. A

reflect radiant heat

Shiny, silver surfaces reflect radiation instead of absorbing it.

5. D

one quarter

Radiation from a small source spreads out as $1/r^2$; doubling r gives ¼.

6. **6.0 × 10³ W**

$P = \dfrac{kA\Delta T}{L} = \dfrac{(0.80)(2.0)(15)}{0.0040} = 6.0 \times 10^3$ W. (Double glazing with an air gap reduces this enormously.)

CHAPTER 4 · LESSON 4.5

The First Law of Thermodynamics

AFTER THIS LESSON YOU CAN

- State the first law and its sign conventions
- Find the work done by a gas from a P–V graph
- Apply the first law to isolated, cyclic, isochoric, isobaric, isothermal and adiabatic processes
- Solve numerical problems with ideal gases

4.5.1 The first law

D FIRST LAW OF THERMODYNAMICS

The change in the internal energy of a system equals the heat added to the system minus the work done by the system (equivalently, plus the work done **on** the system). It is the law of **conservation of energy** for thermodynamic systems.

★ THE TWO SIGN CONVENTIONS (KNOW BOTH!)

Convention	Equation	Meaning of W	Used by
Work done BY the system	$\Delta U = Q - W$	$W > 0$ when the gas **expands**	most exam papers, OpenStax, engineers
Work done ON the system	$\Delta U = Q + W$	$W > 0$ when the gas is **compressed**	the module, Serway
In both: $Q > 0$ when heat **enters** the system; $Q < 0$ when heat **leaves**. The **physics is the same**: compressing a gas or heating it raises U; letting it expand or cool lowers U. Read the question carefully: "the system **does** 100 J of work" means $W_{\text{by}} = +100$ J ($W_{\text{on}} = -100$ J).			

ΔU depends only on the initial and final states (it is a state function), but Q and W separately **depend on the path**.

4.5.2 Work done by a gas

For a gas pushing a piston at pressure P:

$$W_{\text{by}} = \int P\,dV = \text{area under the } P\text{–}V \text{ curve}$$

- Expansion ($\Delta V > 0$): the gas does **positive** work.
- Compression ($\Delta V < 0$): work is done **on** the gas.

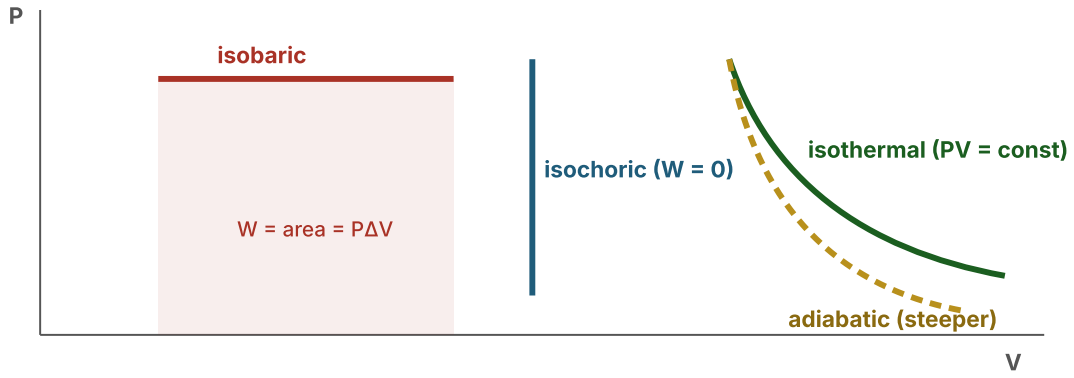
- Constant volume: **no work**.

4.5.3 Special processes

F THE FIRST LAW FOR EACH PROCESS (USING ΔU = Q − W_BY)

Process	What is constant	Key fact	First law becomes
Isolated system	no exchange	$Q = 0$, $W = 0$	$\Delta U = 0$: the internal energy is constant
Cyclic	returns to its start	$\Delta U = 0$	$Q = W$ (net heat in = net work out)
Isochoric (isovolumetric)	volume	$W = 0$	$\Delta U = Q$ (all heat raises U)
Isobaric	pressure	$W = P(V_f - V_i)$	$\Delta U = Q - P\Delta V$
Isothermal (ideal gas)	temperature	$\Delta U = 0$; $W = nRT \ln \frac{V_f}{V_i}$	$Q = W$
Adiabatic	no heat flow	$Q = 0$	$\Delta U = -W$ (expansion cools the gas; compression heats it)

For an ideal **monatomic** gas, $U = \frac{3}{2}nRT$, so $\Delta U = \frac{3}{2}nR\,\Delta T$; $R = 8.314$ J/(mol·K); $PV = nRT$.



Thermodynamic processes on a P–V diagram; the work is the area under each curve

WORKED EXAMPLE 1: HEAT ADDED, WORK DONE (ARBA MINCH EXAM)

400 J of heat is added to a system while it does 100 J of work. Find ΔU.

$\Delta U = Q - W_{\text{by}} = 400 - 100 = +300$ J.

WORKED EXAMPLE 2: HEAT OUT, WORK IN (MODULE, CORRECTED)

2000 J of heat leaves a system and 2500 J of work is done on it. Find ΔU.
$\Delta U = Q + W_{\text{on}} = -2000 + 2500 = +500$ J. (The module's printed working contains arithmetic slips; the correct answer is +500 J.)

✎ WORKED EXAMPLE 3: IDEAL GAS (MODULE)

5000 J of heat is added to 2.00 mol of an ideal monatomic gas at 500 K while the gas does 7500 J of work. Find the final temperature.
$\Delta U = 5000 - 7500 = -2500$ J $= \frac{3}{2}nR\Delta T = \frac{3}{2}(2.00)(8.314)\Delta T \Rightarrow \Delta T = -100$ K → $T_f = 400$ K.
The gas does more work than the heat it receives, so it uses 2500 J of its internal energy and cools.

✎ WORKED EXAMPLE 4: ISOBARIC COMPRESSION (WOLAITA SODO FINAL)

A gas is compressed at a constant 0.800 Pa from 9.00 m³ to 2.00 m³, and 400 J of heat leaves it. Find the work done on the gas and ΔU.
$W_{\text{on}} = -P\Delta V = -(0.800)(2.00 - 9.00) = +5.60$ J.
$\Delta U = Q + W_{\text{on}} = -400 + 5.60 = -394$ J.

✎ WORKED EXAMPLE 5: ISOTHERMAL EXPANSION (MODULE ACTIVITY)

1.0 mol of ideal gas at 0.0 °C expands from 3.0 L to 10.0 L at constant temperature. Find the work done on the gas and the heat exchanged.
$W_{\text{by}} = nRT\ln\frac{V_f}{V_i} = (1.0)(8.314)(273)\ln\frac{10.0}{3.0} = 2.73 \times 10^3$ J, so $W_{\text{on}} = -2.73$ kJ.
Isothermal: $\Delta U = 0 \Rightarrow Q = W_{\text{by}} = +2.73$ kJ (heat flows **into** the gas).

! FIRST-LAW TRAPS

- Decide the sign convention **first** and stick to it.
- **Isothermal** means $\Delta U = 0$, **not** $Q = 0$ (that is **adiabatic**).
- **Isochoric** means $W = 0$.
- In a full cycle $\Delta U = 0$, but Q and W are generally not zero.
- An **isolated** system exchanges neither energy nor matter; a **closed** system exchanges energy but not matter.

Exam Practice

10 QUESTIONS · LESSON 4.5

1 Arba Minch University · Final Exam

In a certain process, 400 J of heat is added to a system and the system simultaneously does 100 J of work. The change in internal energy is:

A) 300 J
B) −500 J
C) −300 J
D) 500 J

2 Wolaita Sodo University · Final Exam 2014

A system absorbs 500 J of heat and does work. If its internal energy increases by 250 J, the work done by the system is:

A) +250 J
B) +750 J
C) −250 J
D) −750 J

3 Wolaita Sodo University · Final Exam 2014

Which of the following characterizes an adiabatic process?

A) $\Delta U = 0$
B) $W = 0$
C) $Q = 0$
D) $\Delta V = 0$

4 Wolaita Sodo University · Final Exam 2014

An isolated system is best described by which statement?

A) Neither matter nor energy can pass into or out of the system
B) Heat can cross the boundary but matter cannot
C) The system has a diathermic boundary
D) The system has reached thermal equilibrium with its surroundings

5 Wolaita Sodo University · Final Exam 2014

Fill in the blank: "The _____ law of thermodynamics states that the change in internal energy of a system equals the net heat flow into the system minus the work done by the system on its surroundings."

6 Wolaita Sodo University · Final Exam 2014

A gas is compressed at a constant pressure of 0.800 Pa from 9.00 m^3 to 2.00 m^3, while 400 J of heat leaves it. Find (a) the work done on the gas and (b) the change in internal energy.

7 Model question (module problem)

A system with an initial internal energy of 205 J absorbs 35 J of heat and does 11 J of work. Find its final internal energy.

8 Model question

In one complete cycle of a heat engine, the change in the internal energy of the working gas is:

A) zero
B) equal to the work done
C) equal to the heat absorbed
D) negative

9 Serway / OpenStax-style problem

An ideal gas expands isothermally. Which statement is true?

A) No heat flows
B) Its internal energy increases
C) No work is done
D) The heat absorbed equals the work done by the gas

10 Model question (module example)

5000 J of heat is added to 2.00 mol of an ideal monatomic gas at 500 K while the gas does 7500 J of work. Find the final temperature.

Answers & Explanations

1. **A**

300 J

$\Delta U = Q - W_{\text{by}} = 400 - 100 = 300$ J.

2. **A**

+250 J

$W_{\text{by}} = Q - \Delta U = 500 - 250 = +250$ J.

3. **C**

Q = 0

Adiabatic means no heat exchange. ΔU = 0 is isothermal (ideal gas); W = 0 and ΔV = 0 are isochoric.

4. **A**

Neither matter nor energy can pass into or out of the system

Isolated: no exchange of energy (heat or work) or matter. B describes a closed system.

5. **First**

$\Delta U = Q - W_{\text{by}}$, conservation of energy.

6. **(a) +5.60 J; (b) −394 J**

(a) $W_{\text{on}} = -P\Delta V = -(0.800)(-7.00) = 5.60$ J.
(b) $\Delta U = Q + W_{\text{on}} = -400 + 5.60 = -394$ J.

7. **229 J**

$\Delta U = 35 - 11 = 24$ J → $U_f = 205 + 24 = 229$ J.

8. **A**

zero

The gas returns to its initial state, and U is a state function, so $\Delta U = 0$ and $Q_{\text{net}} = W_{\text{net}}$.

9. **D**

The heat absorbed equals the work done by the gas

Constant T → $\Delta U = 0$ for an ideal gas, so $Q = W_{\text{by}}$.

10. **400 K**

$\Delta U = -2500$ J $= \frac{3}{2}(2.00)(8.314)\Delta T \Rightarrow \Delta T = -100$ K.

CHAPTER 4 · QUICK REVISION

Heat and Thermodynamics — at a glance

CHAPTER 4 FORMULA SHEET

Topic	Formulas
Temperature	$T_F = \frac{9}{5}T_C + 32$; $T = T_C + 273.15$; $\frac{T_X - X_i}{X_s - X_i} = \frac{T_C}{100}$
Expansion	$\Delta L = \alpha L_0 \Delta T$, $\Delta A = 2\alpha A_0 \Delta T$, $\Delta V = 3\alpha V_0 \Delta T$
Heat	$Q = mc\Delta T$; $C = mc$; $Q = mL$; $\sum Q_{\text{lost}} = \sum Q_{\text{gained}}$
Water	$c = 4186$ J/(kg·°C); $L_f = 3.33 \times 10^5$ J/kg; $L_v = 2.26 \times 10^6$ J/kg
Transfer	conduction $P = kA\Delta T/L$; radiation $P = e\sigma AT^4$
First law	$\Delta U = Q - W_{\text{by}} = Q + W_{\text{on}}$
Gas work	$W_{\text{by}} = P\Delta V$ (isobaric); $nRT\ln(V_f/V_i)$ (isothermal); 0 (isochoric)
Ideal gas	$PV = nRT$; $\Delta U = \frac{3}{2}nR\Delta T$ (monatomic)

THE PROCESS NAMES

Iso-baric = same **bar** (pressure). **Iso-choric** = same **chamber** (volume). **Iso-thermal** = same **therm** (temperature). **A-diabatic** = **no** passing (no heat crosses).